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A novel method for rule extraction in a knowledge-based innovation tutoring system

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ABSTRACT

Symbolic, subsymbolic, and hybrid approaches to rule extraction have so far relied on subsets of first-order logic to cope with the expressiveness trade-off of knowledge representation, on black-box approaches based on artificial neural networks, or on frequent association rule mining in the knowledge discovery and data mining fields. In this article, we present an entirely new method for rule extraction in knowledge-based systems that consists in retrieving an initial set of rules extracted from a knowledge base using conventional logical approaches and then ranking this initial set of rules applying a psychologically motivated multicriteria decision analysis method. We show how this method can be used to implement a knowledge-based management system, demonstrate that this method outperforms the most efficient algorithms for rule extraction proposed to date in the knowledge representation and knowledge discovery fields, and describe its implementation in a knowledge-based innovation tutoring system.

1. Introduction

In the field of Artificial Intelligence there has been a long debate concerning the need for a logical foundation of the knowledge representation language to be used in order to implement commercial-grade, knowledge-based systems. Initial approaches to the problem of knowledge representation postulated a so-called operational semantics, that is, an *ad hoc* semantics not based on formal logics but on the inner workings of the knowledge representation system *per se*, hence the name operational semantics [59,62,64]. Operational semantics was often difficult to formalize and, more importantly, was not well-suited for reasoning in a logically verifiable sense.

During the 1980s and 1990s a discussion ensued in the AI community regarding the need for a logical foundation in a knowledge representation system. As a result, several knowledge representation languages were put forward based on first-order predicate logic [8,9,12]. By the late 1980s, the apparent need for a logical foundation in a knowledge representation system was undisputed in the knowledge representation community [35]. This led to the second – and still unsettled – debate on how much expressiveness was required, that is, how powerful the logical foundation of the knowledge-based system and its associated knowledge representation language should be [80].

With the development of knowledge representation systems in the tradition of KL-ONE [14,18], which were based on the formalism of Semantic Networks [59,84,85] and postulated a subset of first-order predicate logic as the underlying logical foundation of a knowledge representation system, the need for restraining the expressiveness of the underlying semantic representation language became the mainstream approach in the knowledge representation community. This shift toward constraining the expressiveness of the logic was seen as a necessary requirement in order for the underlying inferential algorithms to show good tractability behavior [55]. This trend toward reducing expressiveness in favor of tractability was embraced by most knowledge representation systems based on restricted subsets of first-order predicate logic [3,4,67,83]. As mentioned, this debate has not yet been settled, as there is a difficult trade-off between expressiveness in the knowledge base, on the one hand, and tractability of the underlying inferential algorithms, on the other [88]. This problem of expressiveness is a core problem in the field of knowledge representation, particularly when it comes to the development of knowledge-based systems for real-life applications such as managing processes of innovation and entrepreneurship. Expressiveness in this context is defined as the capability of the knowledge representation language to represent knowledge of the application domain in a way that is computationally tractable and semantically well-defined. Unfortunately, most application domains in the area of management are very knowledge intensive, thus requiring the deployment of large knowledge bases and the use of more expressive knowledge representation languages. As a result, very few knowledge-based

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approaches to evidence-based management have been attempted, often for very restricted application domains, with very few exceptions of large infrastructure projects.

The motivation for the work reported in this article has been the development of efficient algorithms for rule extraction from large knowledge bases to be used in concert with an evidence-based innovation tutoring system designed to bridge the knowledge gaps of emerging regions of innovation and entrepreneurship. The adequacy of such an approach to solving this problem depends upon satisfying the following three conditions [80]:

1. *Knowledge representation*: Defining and implementing a common knowledge representation language and system for representing knowledge and reasoning in a given application domain (innovation and entrepreneurship in knowledge-intensive industries in our case).
2. *Knowledge acquisition*: Acquiring knowledge from a variety of structured and semi-structured sources to generate large knowledge bases containing the knowledge required for a given application domain (the knowledge that is diffused in emerging and complex innovation networks in our case).
3. *Knowledge extraction*: Providing tools for automatic extraction of knowledge rules from knowledge bases so that relevant answers and/or recommendations can be generated and presented to users (rules about processes of innovation and entrepreneurship in our case).

The contribution of this article is a new and highly efficient method to solve the third problem above, namely, the automatic extraction of rules from a knowledge base in knowledge-based management systems. The resulting rule extraction mechanism has been designed to be used in concert with a knowledge-based tutoring system and builds upon a series of functions that measure the relevance of rules in a knowledge base. These relevance functions draw upon a comprehensive body of work on three main fronts: the logical foundations of Artificial Intelligence [8,35], knowledge-based systems and their associated knowledge representation systems[3,4,52–55,80,84–89], and knowledge discovery and data mining [1,50,61,97].

The article is organized as follows. In Section 2, we present previous work in the area of knowledge extraction. In Section 3, we present how the problem of knowledge extraction from large knowledge bases can be modeled as a multicriteria decision analysis problem. In Section 4, we present the multicriteria decision analysis method used to implement the rule extraction mechanism based on the problem definition presented in Section 3. In Section 5, we provide a case that shows how the method for rule extraction introduced in Section 4 is used in a first implementation of a knowledge-based innovation tutoring system so as to comply with the systemic criteria of scalability and performance required for this system. We also present the complexity of our method and show that it outperforms the most efficient algorithms for rule extraction in the fields of knowledge representation and knowledge discovery. In Section 6, we present the discussion of our results. In Section 7, we present our conclusions.

2. Previous work

The problem of knowledge extraction has traditionally been dealt with in the field of knowledge-based systems by choosing less expressive knowledge representation languages. The mainstream approach to building large knowledge representation systems has consisted in reducing the expressiveness of the underlying knowledge representation language so as to be able to implement efficient algorithms for rule extraction and reasoning [55]. Unfortunately, even for highly restricted subsets of first-order predicate logic such as description logics [3,4], terminological logics [14], and typed-feature logics [83], the reasoning algorithms become quite intractable as the number of rules in the knowledge base grow [68].

Conventional approaches to solving this problem have traditionally relied on standard algorithms implemented in knowledge representation systems such as subsumption in description logics [3,4,67] and unification in typed-feature logics [83]. More expressive knowledge representation systems that embrace the full representational power of first-order predicate logic rely on algorithms for forward (input-driven) or backward (goal-driven) inference [80,81]. The latter approaches disregard the expressiveness trade-off of knowledge representation, that is, the fact that more expressive knowledge representations will lead to more inefficient – and even undecidable – algorithms for reasoning. Despite the important initial advances in the area of knowledge extraction, much work still needs to be done in order to build and extract rules from large knowledge bases for commercial-grade applications using symbolic approaches. Large knowledge infrastructure projects such as the CycL project [53,54] demonstrated that this endeavor is indeed possible for commercial-grade applications in specific domains. More recent advances in extracting knowledge from texts in the symbolic tradition such as the work done in connection with the KNEXT project at the University of Rochester promise to provide us with tools for automatic knowledge extraction from semi-structured and noisy text available on the Web today [24–26,42,79,80,82].

In more recent years, the rise of non-symbolic approaches such as artificial neural networks [32,33,78] contributed to the emergence of so-called hybrid approaches to knowledge extraction [44,56,63]. The emergence of these hybrid approaches was fueled by the initial success of connectionist approaches [57] despite the fact that these black-box approaches fail to provide the level of explanation that only white-box symbolic approaches can deliver for complex application domains [15,27,28]. Due to the shortcomings of the rule extraction approaches based on neural networks [47], it became clear that traditional white-box, symbolic approaches based on logics needed to be used in concert with these connectionist, subsymbolic approaches based on neural networks in order to solve the so-called theory and knowledge bottleneck of knowledge-based systems [80]. As a result, further integration of expert systems based on artificial neural networks with other formalisms such as fuzzy logic [99] and rough sets [72] has contributed to the emergence of a wide variety of hybrid expert systems in different application domains [2,19,20,23,45,60].

Closely related with the rule extraction method proposed in this article are a series of methods in the area of association rule mining [1,50,61,97]. These approaches to rule extraction intend to extract a set of nonredundant frequent rules over a given threshold of “interestingness” from a database of transactions consisting of uniquely identified subsets of items taken from a predefined set of items [1]. Frequent association rule mining is an important area of data mining that has suffered from the fact that the association rules mined produced extremely large sets of mostly uninteresting rules [97]. Most of the work in these areas has focused on reducing the search space in order to come up with efficient rule extraction algorithms for complex commercial-grade applications based on the rule extraction algorithms proposed for the less complex basket market analysis applications originally investigated in connection with frequent association rule mining [1,50,61,97]. Our approach also departs from probabilistic approaches in the tradition of belief rule-based systems [21,49,90,103] in that our application domain does not call for heavy-duty inferential processes in the knowledge base. In our application, probabilistic conditionals stored in the knowledge base are used to encode generic knowledge about the world and the application domain of innovation management. They are used to infer new rules over an interestingness threshold and are not intended to implement the kind of predictive models typically associated with belief rule-based systems. Contrary to belief rule-based systems, our application domain requires a more straightforward way of calculating conditional probabilities to efficiently prune the search space of relevant rules, leaving to the user the task of choosing the final rules to be applied.

As we will see in Section 3, the problem we are addressing does not aim to mine association rules from a database of transactions. We assume that a knowledge base is already in place with a sufficiently large amount of nonredundant and interesting rules. We assume that these rules are extracted using hybrid approaches based on the knowledge of experienced innovation managers and on subsymbolic approaches such as those mentioned above. In our application, the challenge consists in deploying a method for extracting the most relevant rules from the knowledge base to answer a user query. The work reported in this article addresses the problem of rule extraction by putting forth an entirely new approach. This new approach may be also classified as a hybrid approach insofar as it makes use of traditional symbolic approaches in the AI logical tradition and also a new family of techniques for rule extraction based on multicriteria decision analysis. The rule extraction problem we are addressing can be stated as follows: Given a query q from a user and a set R of knowledge rules stored in a knowledge base, extract from R the rules that are more relevant in order to come up with an answer to the query q. In order to address this problem as a multicriteria decision analysis problem, we will introduce a set of relevance criteria. The choice of these criteria has been motivated by recent research in the fields of computational linguistics and natural language processing. These criteria are also closely related with metrics used in the field of frequent association rule mining to measure the "interestingness" of a given set of association rules such as the *nonredundancy, productivity*, and *derivability* of rules [96,97].

Our work differs from previous approaches to knowledge extraction in that we put forth a hybrid method for rule extraction that combines a first step based on standard logical approaches with a second step based on multicriteria decision analysis. The second step yields a relevance ranking of the rules extracted from the knowledge base in the first step. The method we propose for rule extraction in large knowledge bases is similar to current approaches to Web search in the field of information retrieval in that we construe rule extraction as a ranking problem based on a relatively small number of relevance criteria but a potentially large number of rules. Unlike methods for frequent rule association that need to extract a set of interesting rules from a database of transactions, the method we have developed needs to extract rules from the knowledge base that have already been entered in a knowledge base and comply with properties of nonredundancy and derivability. Our approach also constitutes quite a departure from conventional symbolic approaches that rely on heavy-duty inferential processes in a knowledge base in that our method deploys multicriteria decision analysis methods to extract the most relevant rules based on a set of relevance criteria. Out of a potentially large set of rules that may match a user query, we build a ranking of relevance with the rules extracted from the knowledge base according to this set of relevance criteria. The rules are then presented to the user following the ranking obtained.

In the next section, we put forth a model describing how rule extraction from large knowledge bases can be approached as a multicriteria decision analysis problem.

3. Modeling rule extraction as a multicriteria decision analysis problem

The problem of rule extraction in knowledge-based systems is of paramount importance in any knowledge-based management system. As we will see in this section, the problem of rule extraction in any knowledge-based management system can be reduced to a ranking problem and can be modeled as a multicriteria decision analysis problem.

3.1. The decision-aiding process

The process of ranking management rules in a knowledge-based management system using multicriteria decision analysis builds upon the process put forth by Bouyssou et al. [13], which we reproduce below.

1. *Problem formulation*: Let Γ be the triplet $\langle R, V, \Pi \rangle$, where R is a set of knowledge rules stored in a knowledge base entailing courses of action recommended by the system to the user, V is a set of points of view and Π is a procedure stating what should be done with the elements of R (which in our case corresponds to extracting the most relevant rules from the knowledge base and ranking them).
2. *Evaluation model*: Let M be the tuple $\langle R, C, U, L \rangle$, where C is a set of relevance criteria derived from V allowing the evaluation of elements of R in terms of each relevance criteria; U is a model of the uncertainty regarding the available information in $R \times C$; and L is an aggregation logic defining the way that the information concerning R and C is operated upon in order to obtain a solution to the problem Π. The evaluation model produces an output, which in our case corresponds to a ranking of the rules extracted from the knowledge base.
3. *Recommendation*: At this third stage, the knowledge-based system generates a recommendation in response to the user query, making sure that the recommendation is given as a ranking of the most relevant rules that answer such query.

The problem of ranking the knowledge-based rules in any knowledge-based system, in general, and in the domain of innovation management and entrepreneurship, in particular, can be modeled as a multicriteria decision analysis problem following the model presented above. As we shall see, we need to put forth a series of relevance functions in order for this problem of automatic rule extraction to be modeled as a multicriteria decision analysis problem. A variety of multicriteria decision analysis methods have been put forth and could be used to assist our user in situations where the task can be defined as a process of making strategic decisions comprising a set of alternatives and a set of criteria. The user is then "guided" to evaluate competing alternatives according to a predefined set of criteria. For the purpose of rule extraction in a knowledge-based system, we construe the set of alternatives as the set of rules that need to be extracted from the knowledge base to answer a user query. This set should contain the rules that are most relevant to answer the query.

3.2. Problem definition

In our application domain, we construe the set of alternatives as the set of knowledge rules $R_q = \{r_1^q, r_2^q, \ldots, r_{n_q}^q\}$ that have been extracted from the knowledge base upon asserting in the knowledge base a query q entered by the user. Using a set of relevance criteria $RC = \{rc_1, rc_2, \ldots, rc_m\}$, the rules in R_q are then ranked and a recommendation is given to the user as a ranking of the rules in R_q. As we shall see in the next subsection, the criteria for assessing relevance in our application domain are "indirectly" defined by the user through his/her query, which differs from conventional multicriteria decision analysis problems in which the set of criteria is usually defined by the decision analyst (the user in our case) or by a panel of experts.

3.3. The relevance functions

Each of the relevance criteria in RC contributes to the relevance of the knowledge rules stored in the knowledge base. In our application domain, though, it is not uncommon that a query q matches the antecedent of a great number of rules in the knowledge base. Therefore, we introduce the main relevance criterion to select the rules that will be part of R_q.

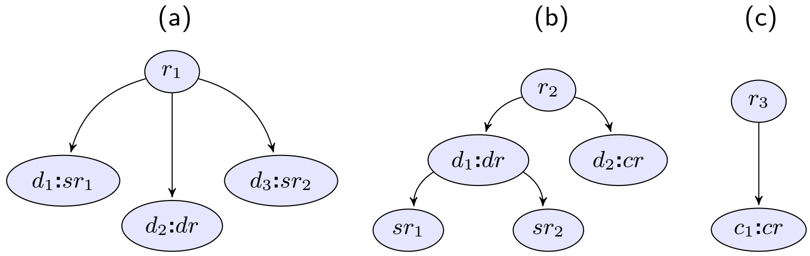

Fig. 1. Examples of simple, conjunctive, disjunctive, and complex rules.

Definition 1. (simple rules) Let e be a region of innovation and entrepreneurship and q a query posed by the user. Let $R_q = \{r_1^q, r_2^q, \ldots, r_{n_q}^q\}$ be the set of n_q knowledge rules extracted whenever the query q is asserted in the knowledge base. A *simple rule* $r_j^q \in R_q$ is defined as a *probabilistic conditional* of the form:

$$r_j^q : \left(a_{r_j^q} \longrightarrow^{p_j^e} c_{r_j^q}\right) \tag{1}$$

The simple rule in (1) is a *probabilistic conditional* stating that whenever $a_{r_j^q}$ occurs in a given region of innovation and entrepreneurship e, then $c_{r_j^q}$ also occurs in e with subjective probability[1] p_j^e. Simple rules are comprised of the *antecedent* $a_{r_j^q}$, the *associated probability* p_j^e, and only one *consequent* $c_{r_j^q}$.

Definition 2. (conjunctive rules) Let e be a region of innovation and entrepreneurship. Let q be a query posed by the user and let $R_q = \{r_1^q, r_2^q, \ldots, r_{n_q}^q\}$ be the set of n_q knowledge rules extracted whenever the query q is asserted in the knowledge base. A *conjunctive rule* r_j^q is defined as a rule of the form:

$$r_j^q : c_1 \wedge \cdots \wedge c_k \wedge \cdots \wedge c_o \tag{2}$$

where c_k, $1 \leq k \leq o$, is a *conjunct* of the form:

$$\begin{aligned} c_k \quad : \quad & \left(a_{r_j^q} \rightarrow^{p_{k,1}^e} c_{r_{k,1}^q}\right) \wedge \cdots \wedge \left(a_{r_j^q} \rightarrow^{p_{k,l}^e} c_{r_{k,l}^q}\right) \\ & \wedge \cdots \wedge \left(a_{r_j^q} \rightarrow^{p_{k,l_k}^e} c_{r_{k,l_k}^q}\right) \end{aligned} \tag{3}$$

The *conjunct* c_k introduces l_k *simple rules*, each one of them expressed as a *probabilistic conditional* introducing one *consequent* $c_{r_{k,l}^q}$ with *associated probability* $p_{k,l}^e$, $1 \leq l \leq l_k$, where l_k is the number of *simple rules* introduced as *consequents* by the *conjunct* c_k. As per our definition, each *conjunct* c_k may introduce a different number l_k of *consequents* as *simple rules*.

Definition 3. (disjunctive rules) Let e be a region of innovation and entrepreneurship. Let q be a query posed by the user and let $R_q = \{r_1^q, r_2^q, \ldots, r_{n_q}^q\}$ be the set of n_q knowledge rules extracted whenever the query q is asserted in the knowledge base. A *disjunctive rule* r_j^q is defined as a rule of the form:

$$r_j^q : d_1 \vee \cdots \vee d_k \vee \cdots \vee d_o \tag{4}$$

where d_k, $1 \leq k \leq o$, is a *disjunct* of the form:

$$\begin{aligned} d_k \quad : \quad & \left(a_{r_j^q} \rightarrow^{p_{k,1}^e} c_{r_{k,1}^q}\right) \vee \cdots \vee \left(a_{r_j^q} \rightarrow^{p_{k,l}^e} c_{r_{k,l}^q}\right) \\ & \vee \cdots \vee \left(a_{r_j^q} \rightarrow^{p_{k,l_k}^e} c_{r_{k,l_k}^q}\right) \end{aligned} \tag{5}$$

The *disjunct* d_k introduces l_k *simple rules*, each one of them expressed as a *probabilistic conditional* introducing one *consequent* $c_{r_{k,l}^q}$ with *associated probability* $p_{k,l}^e$, $1 \leq l \leq l_k$, where l_k is the number of *simple rules* introduced as *consequents* by the *disjunct* d_k. As per our definition, each *disjunct* d_k may introduce a different number l_k of *consequents* as *simple rules*.

Definition 4. (complex rules) Let e be a region of innovation and entrepreneurship. Let q be a query posed by the user and let $R_q = \{r_1^q, r_2^q, \ldots, r_{n_q}^q\}$ be the set of n_q knowledge rules extracted whenever the query q is asserted in the knowledge base. The rule r_j^q, $1 \leq j \leq n_q$, is a *complex rule* if r_j^q is a *disjunctive rule* involving at least one *conjunctive rule* as a *disjunct*.

Example 1. Let us consider the set of knowledge rules K_1 in the knowledge base formed by rules r_1, r_2, and r_3, as shown in Fig. 1, where $d_1, \ldots, d_n$ denote disjuncts of a disjunctive rule and $c_1, \ldots, c_n$ denote conjuncts of a conjunctive rule. In Fig. 1, rule r_1 is a *disjunctive rule* involving the three *disjuncts* d_1, d_2, and d_3, where d_1 and d_3 are *simple rules* and d_2 is a *disjunctive rule*. Since d_2 is a *disjunctive rule*, it follows from the definition of *disjunctive rules* that d_2 does not contain any *conjunctive rules*. Therefore, we conclude that r_1 is not a *complex rule*. Rule r_2 is a *disjunctive rule* involving two *disjuncts*, d_1 and d_2, where d_1 is also a *disjunctive rule* involving two *disjuncts*, both of which are *simple rules* (sr_1 and sr_2), and d_2 is a *conjunctive rule*. Therefore, r_2 corresponds to a *complex rule*. Finally, rule r_3 is comprised of a single *conjunct* c_1 that corresponds to the *conjunctive rule* *cr*. Since c_1 is a *conjunctive rule*, it follows from the definition of *conjunctive rules* that c_1 does not contain any *disjunctive rules*. Therefore, we conclude that r_3 is not a *complex rule*.

Definition 5. (main relevance criterion) Let q be a query posed by the user and r_j^q a *simple rule* or a *conjunctive rule* or a *disjunctive rule* stored in the knowledge base. The rule r_j^q is a *matching rule* to the extent that q matches the *antecedent* of r_j^q.

Applying the main relevance criterion in the knowledge base, we can extract the initial set of matching rules $R_q = \{r_1^q, r_2^q, \ldots, r_{n_q}^q\}$ for a query q entered by the user. This first step will reduce the number of possible rules in the set R_q by extracting from the knowledge base only those rules that are "matched" by the query q. Once the set R_q has been generated, the second step consists in ranking the rules in R_q according to more specific relevance criteria. The relevance criteria used to rank the rules in R_q are presented below. These criteria state that, given a query q, a rule $r_j^q \in R_q$ will be more relevant to the extent that r_j^q triggers more inferences in the knowledge base when q is asserted, to the extent that these inferences are more informative, and to the extent that they are more specific.

Definition 6. (criterion of variety) Given a query q entered by the user and a rule $r_j^q \in R_q$, we define the *variety* v of r_j^q as the number of inferences that can be directly instantiated by r_j^q with *associated probabilities* above α.

Remark. The threshold α provides a lower bound of "interestingness" for the inferences triggered in the knowledge base whenever the antecedent of a probabilistic conditional of the form shown in (1)

[1] This is the subjective probability of experts, that is, the degree of belief provided by experts with world and application domain knowledge.

is matched in the knowledge base. This threshold is referred to as the "interestingness threshold" α [81].[2] Differently from frequent association rule mining approaches that construe the "interestingness" of rules in terms of such criteria as the nonredundancy, derivability, productivity, and closedness of rules [97], we construe interestingness as a subjective probability of experts. As we will show in Section 5, the threshold of "interestingness" will be domain and application-specific. We will show this process by way of introducing the overall architecture of a knowledge-based innovation tutoring system. Rules extracted from the knowledge base will lead to inference chains with a resulting conditional probability. The user will be able to set up this threshold so that only inferences with probabilities over the threshold will be considered and presented to the user.[3]

Lemma 1. Let $r \in R_q$ be a *simple rule*. The *variety* of r has always the value 1 to the extent that its *associated probability* is above the interestingness threshold α and is 0 otherwise.

Proof. A simple rule r instantiates by definition only one inference of form $r : (a \longrightarrow^{p} c)$. It follows immediately that the variety of r has the value 1 if p is above α and 0 otherwise. □

Lemma 2. Let $r \in R_q$ be a *disjunctive rule*. The *variety* of r has always the value that results of adding all the *consequents* introduced by the *disjuncts* in r with *associated probabilities* above the interestingness threshold α.

Proof. Let r be a disjunctive rule r: $d_1 \vee \cdots \vee d_k \vee \cdots \vee d_o$, where the disjunct d_k, $1 \le k \le o$, has the form:

$$d_k : (a \longrightarrow^{p_{k,1}} c_{k,1}) \vee \cdots \vee (a \longrightarrow^{p_{k,l_k}} c_{k,l_k}) \tag{6}$$

where l_k is the number of disjuncts in d_k. By definition, we have that any disjunctive rule in R_q has the same antecedent a matched by the query q posed by the user. It follows that r can be rewritten as the disjunctive rule involving $\sum_{k=1}^{o} l_k$ simple rules, as shown below:

$$r : a \longrightarrow \begin{cases} c_{1,1}^{p_{1,1}} \vee \cdots \vee c_{1,l}{}^{p_{1,l}} \vee \cdots \vee c_{1,l_1}{}^{p_{1,l_1}} \vee \\ \cdots \\ c_{k,1}{}^{p_{k,1}} \vee \cdots \vee c_{k,l}{}^{p_{k,l}} \vee \cdots \vee c_{k,l_k}{}^{p_{k,l_k}} \vee \\ \cdots \\ c_{o,1}{}^{p_{o,1}} \vee \cdots \vee c_{o,l}{}^{p_{o,l}} \vee \cdots \vee c_{o,l_o}{}^{p_{o,l_o}} \end{cases} \tag{7}$$

Therefore, rule r can be rewritten as shown in (8), which involves $\sum_{k=1}^{o} l_k$ simple rules (each one of them introducing only one consequent):

$$r : \left(a \longrightarrow \bigvee_{k=1}^{o} \bigvee_{l=1}^{l_k} c_{k,l}{}^{p_{k,l}} \right) \tag{8}$$

Rule r instantiates $\sum_{k=1}^{o} l_k$ inferences in the knowledge base, each one of them associated with a simple rule of form $(a \longrightarrow^{p_{k,l}} c_{k,l})$, where $1 \le k \le o$ and $1 \le l \le l_k$. The variety of r for a given interestingness threshold α corresponds to $v = \sum_{k=1}^{o} \sum_{l=1}^{l_k} v_{k,l}$, where $v_{k,l} = 1$ if $p_{k,\,l}$ is above α and 0 otherwise. As a corollary, it follows that the variety v of r is such that $0 \le v \le \sum_{k=1}^{o} l_k$. □

Lemma 3. Let $r \in R_q$ be a *conjunctive rule*. The *variety* of r has always the value 1 to the extent that all the *conjuncts* in r_j^q introduce *consequents* with *associated probabilities* such that the resulting conditional probability is above the interestingness threshold α and 0 otherwise.

Proof. Let r be a conjunctive rule of the form:

$$r : c_1 \wedge \cdots \wedge c_k \wedge \cdots \wedge c_o \tag{9}$$

where the conjunct c_k, $1 \le k \le o$, has the form:

$$c_k : (a \longrightarrow^{p_{k,1}} c_{k,1}) \wedge \cdots \wedge (a \longrightarrow^{p_{k,l_k}} c_{k,l_k}) \tag{10}$$

and where l_k is the number of conjuncts in c_k. By definition, we have that any conjunctive rule in R_q has the same antecedent a matched by the query q posed by the user. It follows that r can be rewritten as the conjunctive rule involving $\sum_{k=1}^{o} l_k$ simple rules, as shown below:

$$r : a \longrightarrow \begin{cases} c_{1,1}^{p_{1,1}} \wedge \cdots \wedge c_{1,l}{}^{p_{1,l}} \wedge \cdots \wedge c_{1,l_1}{}^{p_{1,l_1}} \wedge \\ \cdots \\ c_{k,1}{}^{p_{k,1}} \wedge \cdots \wedge c_{k,l}{}^{p_{k,l}} \wedge \cdots \wedge c_{k,l_k}{}^{p_{k,l_k}} \wedge \\ \cdots \\ c_{o,1}{}^{p_{o,1}} \wedge \cdots \wedge c_{o,l}{}^{p_{o,l}} \wedge \cdots \wedge c_{o,l_o}{}^{p_{o,l_o}} \end{cases} \tag{11}$$

Rule r can be rewritten as a rule involving only simple rules, each one of them introducing just one consequent, as shown in (12):

$$r : \left(a \longrightarrow \bigwedge_{k=1}^{o} \bigwedge_{l=1}^{l_k} c_{k,l}{}^{p_{k,l}} \right) \tag{12}$$

Therefore, the consequents $c_{k,\,l}$ in (12) can be reduced to only one consequent of form $c_{1,\,1} \wedge \cdots \wedge c_{o,l_o}$. It follows that only one inference of the form:

$$r : (a \longrightarrow^{p} c_{1,1} \wedge \cdots \wedge c_{o,l_o}) \tag{13}$$

can be instantiated if and only if $p = \prod_{k=1}^{o} \prod_{l=1}^{l_k} p_{k,l} > \alpha$, where p is the conditional probability that results from all associated probabilities $p_{k,l}$. We conclude that the variety of r for a given interestingness threshold α corresponds to the value 1 if all simple rules $(a \longrightarrow^{p_{k,l}} c_{k,l})$ in (12) are such that the resulting conditional probability p is above α. Otherwise, the variety of r is 0. □

Lemma 4. Let $r \in R_q$ be a *complex rule*. The *variety* of r corresponds to the sum of the *variety* of all *disjuncts* in r that are *disjunctive rules* plus the number of *disjuncts* in r that are *conjunctive rules* with a *variety* equal to 1.

Proof. Let r be a complex rule of form r: $d_1 \vee \cdots \vee d_k \vee \cdots \vee d_o$, $1 \le k \le o$, where at least the disjunct d_k is a conjunctive rule. As per Lemma 3, any disjunct d_k that is a conjunctive rule can be expressed as:

$$d_k : \left(a \longrightarrow \bigwedge_{k=1}^{l_k} c_{k,l}{}^{p_{k,l}} \right) \tag{14}$$

where $1 \le l \le l_k$ and l_k is the number of conjuncts in the conjunctive rule d_k. As per Lemma 3, any disjunct d_k corresponding to a conjunctive rule can be expressed as in (14) and will contribute the value 1 or 0 to the variety of r. As per Lemma 2, the variety of all disjuncts that are disjunctive rules in r for a given interestingness threshold α will correspond to the total number of consequents introduced by them such that their associated probabilities $p_{k,\,l}$ are above α. Therefore, the variety of r corresponds to the sum of the variety of all disjuncts in r that are disjunctive rules with a variety equal to 1, that is, with associated probability above the interestingness threshold α, plus the number of disjuncts in r that are conjunctive rules with a variety equal to 1. □

Definition 7. We define the function $variety(r_j^q, \alpha)$ as a function that maps rules in R_q to $v \in \mathbb{Z}_{\ge 0}$ such that $0 \le v \le \sum_{k=1}^{o} l_k$. The higher the resulting value of v, the higher the relevance this variety criterion contributes to r_j^q.

Example 2. Let us consider the set of knowledge rules K_2 in the knowledge base formed by rules r_1, r_2, and r_3, as shown in Fig. 2,

[2] Our interestingness threshold α differs from the one introduced by Schubert and Hwang [81] in that we construe α as a subjective rather than a statistical probability.

[3] As shown by other authors, most of the "world knowledge" that needs to be represented in a knowledge-based system corresponds to probabilistic conditionals and factoids [5,80]. Probabilistic conditionals will lead to inference chains with conditional probabilities attached to them via conventional forward-chaining inference mechanisms.

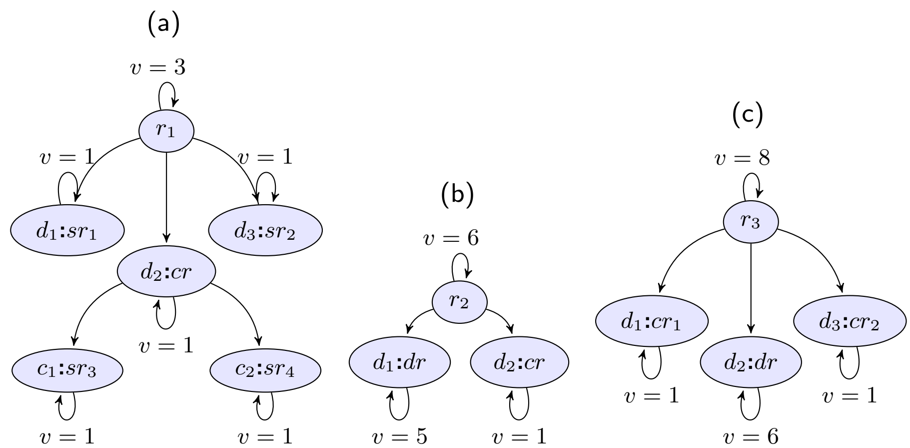

Fig. 2. Examples of rules and their respective varieties.

where we assume that all *associated probabilities* are above α. In Fig. 2, $d_1, \dots, d_n$ are *disjuncts* and $c_1, \dots, c_n$ are *conjuncts*, and *sr, dr*, and *cr* stand for *simple rule, disjunctive rule*, and *conjunctive rule*, respectively. The *variety* of the rules in Fig. 2 can be calculated as follows:

$$\begin{aligned} variety(r_1, \alpha) &= variety(d_1, \alpha) + variety(d_2, \alpha) + variety(d_3, \alpha) \\ &= variety(sr_1, \alpha) + variety(cr, \alpha) + variety(sr_2, \alpha) \\ &= 3 \\ variety(r_2, \alpha) &= variety(d_1, \alpha) + variety(d_2, \alpha) \\ &= variety(dr, \alpha) + variety(cr, \alpha) \\ &= 6 \\ variety(r_3, \alpha) &= variety(d_1, \alpha) + variety(d_2, \alpha) + variety(d_3, \alpha) \\ &= variety(cr_1, \alpha) + variety(dr, \alpha) + variety(cr_2, \alpha) \\ &= 8 \end{aligned}$$

Example 3. If a query q matches the antecedent a of a disjunctive rule r of the form r: $(a \longrightarrow^{p_1} c_1) \vee (a \longrightarrow^{p_2} c_2)$, then there are two inferences instantiated by r whenever q is asserted in the knowledge base. If the defined interestingness threshold α were set to 0.5 and p_1 and p_2 were 0.7 and 0.45, respectively, the variety v of r would be $v = 1$ because p_2 would be below α. If α were lowered to, say, 0.4, the variety of r would be $v = 2$ because there would be two inferences instantiated by r above α whenever q is asserted in the knowledge base.

Example 4. If a query q matches the antecedent a of a conjunctive rule r of the form r: $(a \longrightarrow^{p_1} c_1) \wedge (a \longrightarrow^{p_2} c_2)$, then there is only one inference instantiated by r whenever q is asserted in the knowledge base. Assuming that the defined interestingness threshold α is 0.5 and that p_1 and p_2 are 0.7 and 0.45, respectively, then the variety v of r would be $v = 0$ because the resulting conditional probability $p = p_1 \times p_2 = 0.315$ would be below α. If α were lowered to, say, 0.3, then the variety of r_1 would be $v = 1$ because there would be one inference instantiated by r whenever the query q is asserted in the knowledge base of form r: $(a \longrightarrow^{p} (c_1 \wedge c_2))$ such that $p = p_1 \times p_2$ is above α.

Definition 8. (criterion of informativeness) Let q be a query entered by the user. Let $r_j^q \in R_q$ be a rule in the knowledge base with *antecedent* $a_{r_j^q}$ such that $1 \le j \le n_q$. Let $C = \{c_1, \dots, c_n\}$ be the set of consequents immediately instantiated by r_j^q whenever q is asserted in the knowledge base. We define the *informativeness* of c_i as the number that results of adding the total number of *consequents* introduced in c_i with conditional probabilities above the probability threshold α. Let $I = \{i_1, \dots, i_n\}$ be the set containing the *informativeness* values of all consequents in C. We define the *informativeness* i of r_j^q as $i = max(I)$. Therefore, i will correspond to the *informativeness* of the consequents in C with the highest *informativeness* value.

Lemma 5. Let $r \in R_q$ be a simple rule. The *informativeness* of r has always the value 1 to the extent that its *associated probability* is above the interestingness threshold α and is 0 otherwise.

Proof. A simple rule r of form r: $(a \longrightarrow^{p} c)$ introduces by definition only one consequent and initiates only one inference. It follows immediately that the informativeness of r has the value 1 if p is above α and 0 otherwise. □

Lemma 6. Let $r \in R_q$ be a *disjunctive rule*. The *informativeness* of r has always the value 1 to the extent that at least one of the *disjuncts* in r introduces at least one *consequent* with *associated probability* p above the interestingness threshold α and is 0 otherwise.

Proof. Let r be a disjunctive rule r: $d_1 \vee \cdots \vee d_k \vee \cdots \vee d_o$, where the disjunct d_k, $1 \le k \le o$, has the form:

$$d_k : (a \longrightarrow^{p_{k,1}} c_{k,1}) \vee \cdots \vee (a \longrightarrow^{p_{k,l_k}} c_{k,l_k}) \tag{15}$$

where l_k is the number of disjuncts in d_k. By definition, we have that any disjunctive rule in R_q has the same antecedent a matched by the query q posed by the user. It follows that r can be rewritten as the disjunctive rule involving $\sum_{k=1}^{o} l_k$ simple rules, as shown below:

$$r : a \longrightarrow \begin{cases} c_{1,1}^{p_{1,1}} \vee \cdots \vee c_{1,l}{}^{p_{1,l}} \vee \cdots \vee c_{1,l_1}{}^{p_{1,l_1}} \vee \\ \cdots \\ c_{k,1}{}^{p_{k,1}} \vee \cdots \vee c_{k,l}{}^{p_{k,l}} \vee \cdots \vee c_{k,l_k}{}^{p_{k,l_k}} \vee \\ \cdots \\ c_{o,1}{}^{p_{o,1}} \vee \cdots \vee c_{o,l}{}^{p_{o,l}} \vee \cdots \vee c_{o,l_o}{}^{p_{o,l_o}} \end{cases} \tag{16}$$

Therefore, rule r can be rewritten as shown in (17), which involves $\sum_{k=1}^{o} l_k$ simple rules (each one of them introducing only one consequent):

$$r : \left(a \longrightarrow \bigvee_{k=1}^{o} \bigvee_{l=1}^{l_k} c_{k,l}{}^{p_{k,l}} \right) \tag{17}$$

It follows that rule r instantiates $\sum_{k=1}^{o} l_k$ inferences in the knowledge base, each one of them associated with a simple rule that introduces just one consequent and has the form $(a \longrightarrow^{p_{k,l}} c_{k,l})$, where $1 \le k \le o$ and $1 \le l \le l_k$. Therefore, the informativeness of these rules is either 1 or 0 depending on whether there is at least one associated probabilities $p_{k,\,l}$ above α or not. It follows that the informativeness of r is $i = max(I) = 1$ if and only if there is at least one simple rule in (16) such that $p_{k,\,l}$ is above α and 0 otherwise. As a corollary, it follows that the informativeness i of r is such that $0 \le i \le 1$. □

Lemma 7. Let $r \in R_q$ be a *conjunctive rule*. The *informativeness* of r has the value that results of adding all the *consequents* introduced by all the *conjuncts* of r if the probabilities associated with all the *consequents* introduced by all the *conjuncts* in r are such that the resulting conditional probability is above the interestingness threshold α and is 0 otherwise.

Proof. Let r be a conjunctive rule of form:

$$r : c_1 \wedge \cdots \wedge c_k \wedge \cdots \wedge c_o \tag{18}$$

where the conjunct c_k, $1 \le k \le o$, has the form:

$$c_k : (a \longrightarrow^{p_{k,1}} c_{k,1}) \wedge \cdots \wedge (a \longrightarrow^{p_{k,l_k}} c_{k,l_k}) \tag{19}$$

where l_k is the number of conjuncts in c_k. By definition, we have that any conjunctive rule in R_q has the same antecedent a matched by the query q posed by the user. It follows that r can be rewritten as the conjunctive rule involving $\sum_{k=1}^{o} l_k$ simple rules, as shown below:

$$r : a \longrightarrow \begin{cases} c_{1,1}^{p_{1,1}} \wedge \cdots \wedge c_{1,l}{}^{p_{1,l}} \wedge \cdots \wedge c_{1,l_1}{}^{p_{1,l_1}} \wedge \\ \cdots \\ c_{k,1}{}^{p_{k,1}} \wedge \cdots \wedge c_{k,l}{}^{p_{k,l}} \wedge \cdots \wedge c_{k,l_k}{}^{p_{k,l_k}} \wedge \\ \cdots \\ c_{o,1}{}^{p_{o,1}} \wedge \cdots \wedge c_{o,l}{}^{p_{o,l}} \wedge \cdots \wedge c_{o,l_o}{}^{p_{o,l_o}} \end{cases} \tag{20}$$

Rule r can be rewritten as a rule involving only simple rules, each one of them introducing just one consequent, as shown in (21):

$$r : \left(a \longrightarrow \bigwedge_{k=1}^{o} \bigwedge_{l=1}^{l_k} c_{k,l}{}^{p_{k,l}} \right) \tag{21}$$

Therefore, the consequents $c_{k,l}$ in (21) can be reduced to only one consequent of form $c_{1,1} \wedge \cdots \wedge c_{o,l_o}$. It follows that only one inference of the form:

$$r : (a \longrightarrow^{p} c_{1,1} \wedge \cdots \wedge c_{o,l_o}) \tag{22}$$

can be instantiated if and only if the conditional probability $p = \prod_{k=1}^{o} \prod_{l=1}^{l_k} p_{k,l}$ that results from all associated probabilities $p_{k,l}$ is above α. We conclude that the set C is such that $|C| = 1$ and $I = \{i\}$ where i is such that:

$$i = \begin{cases} \sum_{k=1}^{o} l_k \iff & (\prod_{k=1}^{o} \prod_{l=1}^{l_k} p_{k,l}) > \alpha \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

As a corollary, we have that if r carries any informativeness value, then this value is always $\sum_{k=1}^{o} l_k$. $\square$

Lemma 8. Let $r \in R_q$ be a *complex rule*. The *informativeness* of r corresponds to the *informativeness* of its conjunctive part if there is at least one conjunct in the conjunctive part of r with an *informativeness* value greater than or equal to 1 and will correspond to the *informativeness* of the *conjunct* with the highest *informativeness* value. Otherwise, it corresponds to the *informativeness* of its disjunctive part and will have either the value 1 or 0.

Proof. Let r be a complex rule of form *r:* $d_1 \vee \cdots \vee d_k \vee \cdots \vee d_o$, where $1 \le k \le o$ and at least the disjunct d_k corresponds to a conjunctive rule. As per Lemma 3, we can express any disjunct d_k corresponding to a conjunctive rule as the following rule:

$$d_k : \left(a \longrightarrow \bigwedge_{k=1}^{l_k} c_{k,l}{}^{p_{k,l}} \right) \tag{24}$$

where l_k is the number of conjuncts in the conjunctive rule d_k. As per Lemma 7, the informativeness contributed to r by its conjunctive part will correspond to the informativeness of the conjunct with the highest informativeness value in this conjunctive part, where the informativeness value of each of these conjuncts will be either 0 or $\sum_{k=1}^{o} l_k$ depending on whether the conjunct introduces consequents such that the resulting conditional probability p is below α or not. As per Lemma 6, the informativeness of all disjuncts that are disjunctive rules in r for a given interestingness threshold α will correspond to the value 1 or 0 depending on whether or not their associated probabilities $p_{k,l}$ are above α. Consequently, the total informativeness contributed to r by its disjunctive part will have the value 1 or 0. Therefore, the informativeness of the conjunctive part of r will always correspond to the informativeness of its conjunctive part if there is at least one conjunct in the conjunctive part of r with an informativeness value equal or higher to 1 and will otherwise correspond to the informativeness value of its disjunctive part and will have the value either 1 or 0. $\square$

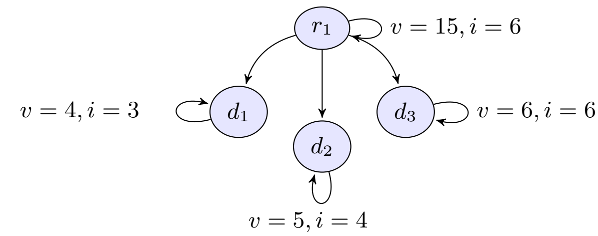

Fig. 3. Example of complex rules.

Example 5. Let us consider the set of knowledge rules K_2 in Fig. 3, where $d_1, \ldots, d_n$ are *disjuncts* and all *associated probabilities* are above the interestingness threshold α. Rule r_1 is a complex rule consisting of the disjuncts d_1, d_2, and d_3. Rule r_1 has a variety value of 15 (the sum of the variety of its disjuncts) and an informativeness value of 6 (the highest informativeness value among its disjuncts). The disjuncts d_1, d_2, and d_3 in Fig. 3 are complex rules, as indicated by their respective informativeness values.[4]

Definition 9. We introduce the function $informativeness\ (r_j^q, \alpha)$ as a function that takes rules in R_q as arguments and maps them to their *informativeness* value $i \in \mathbb{Z}_{\ge 0}$. The higher the resulting value of the *informativeness* i, the higher the relevance this *informativeness* criterion contributes to r_j^q.

Example 6. The *informativeness* of the rules in Fig. 3 can be calculated as follows (assuming that all the *associated probabilities* are above α):

$$\begin{aligned} informativeness(r_1, \alpha) = max(&informativeness(d_1, \alpha), \\ &informativeness(d_2, \alpha), \\ &informativeness(d_3, \alpha)) \\ = 6& \end{aligned}$$

Example 7. If a query q matches the *antecedent* a_1 of a *conjunctive rule* r_1 of the form $r_1 : (a_1 \longrightarrow^{p_1} b_1) \wedge (a_1 \longrightarrow^{p_2} b_2)$, then there are two *consequents* introduced by r_1 whenever q is asserted in the knowledge base. Assuming that the defined interestingness threshold α is 0.5 and that p_1 and p_2 are 0.7 and 0.4, respectively, then the *informativeness* i of r_1 would be $i = 0$ since the resulting conditional probability would be $p = 0.28$, which is below α. If α were lowered to, say, 0.25, then the *informativeness* of r_1 would be $i = 2$ as there would be two *consequents* triggered by r_1 when q is asserted in the knowledge base and the resulting conditional probability would be above α.

Definition 10. (criterion of specificity) Let q be a query posed by the user and $r_j^q \in R_q$ a rule in the knowledge base with *antecedent* $a_{r_j^q}$, $1 \le$

[4] The fact that these disjuncts have an informativeness value greater than 1 indicates that they include conjunctive rules. Otherwise, they would have the informativeness value 1 as per Lemma 6 because their associated probabilities are above α.

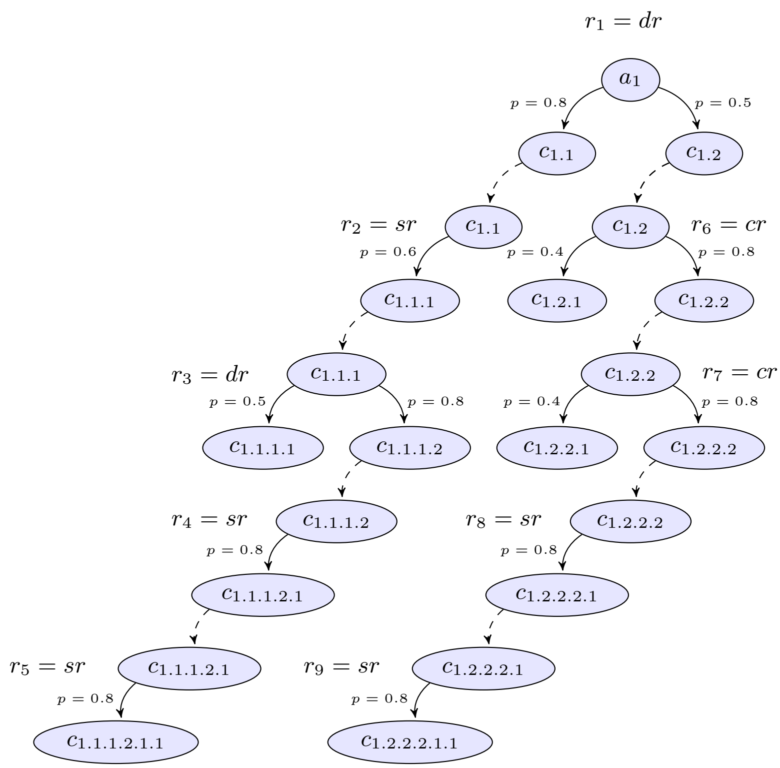

Fig. 4. Example of inference chains in the knowledge base.

$j \le n_q$. Let $IC = \{ic_1, \ldots, ic_n\}$ be the set of all inference chains triggered from r_j^q whenever q is asserted in the knowledge base such that the resulting conditional probability of the last inference in the inference chains in IC is above α. Let l_i be the path length of inference chain $ic_i \in IC, 1 \le i \le n$, and let $L = \{l_1, \ldots, l_n\}$ be the set of all path lengths associated with the inference chains in IC. We define the *specificity* s of r_j^q as $s = max(L)$. Thus, the *specificity* of r_j^q corresponds to the path length of the inference chain in IC with the longest path.

Example 8. Let us assume that q matches the antecedent $a_{r_1^q}$ of the following *simple rule* $r_1^q : a_{r_1^q} \longrightarrow^{p_1=0.7} c_{r_1^q}$. Let us assume further that the *consequent* $c_{r_1^q}$ matches the *antecedent* of the following *simple rule* $r_2^q : c_{r_1^q} \longrightarrow^{p_2=0.5} c_{r_2^q}$. Let us assume further that the *consequent* $c_{r_2^q}$ matches the *antecedent* of the following *simple rule* $r_3^q : c_{r_2^q} \longrightarrow^{p_3=0.4} c_{r_3^q}$. Let us assume further that the interestingness threshold α has been set to 0.30 and that there are no other rules in the knowledge base that can be triggered whenever q is asserted in the knowledge base. As a result, there would be only one inference chain $ic_1 \in IC$ triggered by r_1^q whenever q is asserted in the knowledge base, namely:

$$ic_1 : a_{r_1^q} \overset{inf1}{\longmapsto} c_{r_1^q} \overset{inf2}{\longmapsto} c_{r_2^q} \overset{inf3}{\longmapsto} c_{r_3^q}$$

Consequently, $IC = \{ic_1\}$. Although the path length l_1 of this inference chain is $l_1 = 3$, the set L corresponds to $L = \{l_1\}$, with $l_1 = 2$, because the conditional probabilities associated with this inference chain would be $cp(inf1) = 0.7$, $cp(inf2) = 0.35$, and $cp(inf3) = 0.14$. The last inference $inf3$ in this inference chain would be below the interestingness threshold α. According to the definition of *specificity*, this last inference is disregarded and the path length of ic_1 corresponds to $l_1 = 2$ instead of $l_1 = 3$. It follows that the *specificity* of r_1^q is $s = max(L) = 2$. If the interestingness threshold α were lowered to, say, 0.1, then the *specificity* of r_1^q would be $s = max(L) = 3$ because the entire inference chain would now be above the interestingness threshold α.

Example 9. Let us consider the rules $r_1, \ldots, r_9$ in Fig. 4, where *sr, cr*, and *dr* stand for *simple rule, conjunctive rule* and *disjunctive rule*, respectively. In Fig. 4, *a* and *c* stand for *antecedent* and *consequent*, respectively, and the dashed arrows denote rules in the knowledge base matching *antecedents* in an inference chain. Let us further consider that the interestingness threshold α has been set to 0.25. As we can see, the *specificity* of rule r_1 corresponds to $s = 4$ because we have two inference chains of length 4 above the interestingness threshold α. The *specificity* criterion would assign in this case the value $s = 4$ to rule r_1 whenever a query q is asserted in the knowledge base. This would be equivalent to stating that the most specific information that can be gathered from the knowledge base whenever q is asserted can be represented by the rule:

$$r_1 : \begin{cases} a_1 \longrightarrow^{p_1=0.3072} (c_{1.1} \wedge c_{1.1.1} \wedge c_{1.1.1.2} \wedge c_{1.1.1.2.1}) \vee \\ a_1 \longrightarrow^{p_2=0.256} (c_{1.2} \wedge c_{1.2.2} \wedge c_{1.2.2.2} \wedge c_{1.2.2.2.1}) \end{cases} \tag{25}$$

Definition 11. We introduce the function $specificity(r_j^q, \alpha)$ as a function that takes rules in R_q as arguments and maps them to their specificity value $s \in \mathbb{Z}_{\ge 0}$. The higher the resulting value of s, the higher the relevance this specificity criterion contributes to r_j^q.

Remark. There is a fundamental difference between the informativeness and the specificity criteria. While the informativeness of a rule r gives us the amount of information it contributes in terms of its number of consequents, the specificity of a rule involves more rules with their respective and different antecedents in an inference chain. Therefore, the specificity criterion provides more specific information. Furthermore, the specificity criterion is sensitive to the context of a rule in the knowledge base as the value of the specificity criterion would depend not only on rule r but also on other rules and facts stored in the knowledge base that can provide more specific information along a given inference chain.

3.4. The relevance ranking

While the rules in R_q can only be defined once the user has entered a query q, the set of relevance criteria $RC = \{rc_1, \ldots, rc_m\}$, with $1 \leq i \leq m = 3$, is independent of the query q and is therefore a fixed set of criteria according to which we can rank the rules in R_q. Once the set R_q has been determined for a query q, the relevance ranking is built using the set of relevance criteria in RC, which correspond to the *variety*, *informativeness*, and *specificity* criteria introduced above. As we will see in the next section, this ranking will ultimately depend on the multicriteria decision analysis method used.

In the next section, we motivate and introduce the use of TODIM as a multicriteria decision analysis method for calculating this ranking and explain how this ranking is built.

4. Multicriteria decision analysis methods

Several discrete multicriteria decision analysis methods have been put forth to solve complex decision-making problems [6,7,36,92]. In this section, we showcase the use of multicriteria decision analysis methods to solve the ranking problem in our application domain using the TODIM multicriteria decision analysis method, as introduced in [40].

4.1. The TODIM method

TODIM was the first multicriteria decision analysis method to deploy a value function based on Prospect Theory [48,91]. Since its inception in the early 1990s [40], the TODIM method has been applied to deal with complex multicriteria decision analysis problems in a wide variety of application domains such as real estate evaluation [38,39,65,66], business planning [74,75], energy resource management [37,39], and hospital management [37,39]. TODIM has also been extended to deal with either hybrid or fuzzy data [19,29,51,58,73,102]. We refer the reader to [29,40,51] for a more detailed description of the TODIM method.

4.2. The rationale for using the TODIM method

TODIM, the multicriteria decision method we have chosen to be used in concert with the multicriteria decision analysis model introduced in Section 3, was the first to introduce a psychological account of how human evaluators make decisions under risk. Conventional multicriteria decision analysis methods were originally based on utility theory [69] and did not take into account the way individuals go about making decisions under risk. The work on Prospect Theory and Cumulative Prospect Theory [48,91] demonstrated that individuals show a great deal of risk aversion when they face decisions under risk involving gains and, more interestingly, that they show risk-seeking behavior when they face losses. This distinction between individuals making decisions based on the gains or losses of competing alternatives, as opposed to the final asset positions entailed by the alternatives they face, is one of the key findings of Prospect Theory and Cumulative Prospect Theory, which is built into the value function of the TODIM method.

TODIM was the first multicriteria decision analysis method to incorporate this important finding in its value function, thus providing a psychologically motivated method for strategic decision-making under risk. The risk-seeking behavior predicted by Prospect Theory and Cumulative Prospect Theory is implemented in TODIM by way of introducing a so-called factor of attenuation of losses. This factor "attenuates" the losses in the convex (negative) part of the value function used by TODIM so that this function gets steeper in its negative part, thus accounting for the risk-seeking behavior of decision makers when losses arise. The use we are making of TODIM for the purposes of evaluating the relevance of rules in a knowledge base is based on the need to make such process comply with the way individuals, including experts, make decisions under risk in the real world. This will lead to recommendations that are more attuned to the perception of risk and the attitudes toward decision-making under risk of individual managers and experts. This is particularly the case in the area of innovation management due to the inherent uncertainty associated with innovation processes. We elaborate further on this important point in Section 5.

4.3. Applying the TODIM method

Let us consider $R_q = \{r_1^q, \ldots, r_j^q, \ldots, r_{n_q}^q\}$, the set of n_q knowledge rules (usually referred to as alternatives in the multicriteria decision analysis literature) to be ordered, and $RC = \{rc_1, \ldots, rc_i, \ldots, rc_m\}$, the set of m relevance criteria (usually referred to as criteria in the multicriteria decision analysis literature). The evaluation of the rules in relation to all relevance criteria in RC is a matrix of evaluation. This matrix contains values that are all numerical. Let us define $S = [S_{nm}]$, where $S_{nm} \in [0, 1]$, as the matrix of normalized rules' scores against relevance criteria. We then define the importance of each relevance criterion, expressed as w_i. We also introduce a reference relevance criterion as the one with the highest importance. Let rc_r be the reference relevance criterion, whose importance is defined as w_{rc}. Now we can compute the relative weight of each criterion in RC as:

$$w_{ir} = w_i / w_{rc} \tag{26}$$

Once the relative weight for each criterion has been elicited, the following value function is defined:

$$\phi_i(S_j, S_k) = \begin{cases} \sqrt{\dfrac{w_{ir}(S_{ji} - S_{ki})}{\sum_{i=1}^{m} w_{ir}}} & if\ (S_{ji} - S_{ki}) > 0 \\ 0 & if\ (S_{ji} - S_{ki}) = 0 \\ -\dfrac{1}{\vartheta}\sqrt{\dfrac{\sum_{i=1}^{m} w_{ir}(S_{ki} - S_{ji})}{w_{ir}}} & if\ (S_{ji} - S_{ki}) < 0 \end{cases} \tag{27}$$

This is a pairwise comparison function modeling the preferences that the innovation tutoring system will provide to users. When $(S_{ji} - S_{ki})$ is positive, gains are experienced and the concave form of the value function denotes the aversion to risk by the system. For negative differences, losses arise. In this case, the value function ϕ_i will adopt a convex form, according to Prospect Theory and Cumulative Prospect Theory [48,91]. In this case, TODIM introduces the factor of attenuation of losses ϑ in order to account for one of the main findings of Prospect Theory and Cumulative Prospect Theory, namely, that individuals show a propensity to risk when losses arises. Thus, the factor ϑ will make the value function adopt a steeper form nearer the reference point, a point that is defined in Prospect Theory and Cumulative Prospect Theory as that which represents the *status quo* or the current asset position of evaluators, that is, the asset position before gains or losses arise. In Section 5, we elaborate on how to this factor of attenuation of losses is handled in TODIM.

TODIM then calculates the global dominance relation between each pair of rules (S_j, S_k), as shown in (28):

$$\delta(S_j, S_k) = \sum_{i=1}^{m} \phi_i(S_j, S_k) \tag{28}$$

Finally, TODIM aggregates the values $\delta(S_j, S_k)$ to obtain a global dominance value for each rule in the following way:

$$\xi_j = \frac{\sum_{k=1}^{n_q} \delta(S_j, S_k) - min_j \sum_{k=1}^{n_q} \delta(S_j, S_k)}{max_j \sum_{k=1}^{n_q} \delta(S_j, S_k) - min_j \sum_{k=1}^{n_q} \delta(S_j, S_k)} \tag{29}$$

Expression (29) represents a normalized global relevance of rule r_j^q when compared to all other rules in R_q. Determining ξ_j leads to the ordering, or ranking, of the rules in R_q.

4.4. Rule extraction using TODIM

As stated in Sections 1 and 2, the problem of rule extraction has traditionally been one of the main obstacles for the rapid adoption of knowledge-based approaches to implementing commercial-grade enterprise software applications in a variety of application domains. Unfortunately, innovation management is no exception. The multicriteria decision analysis method introduced above provides us with a tool for implementing rule extraction from large knowledge bases. The function ξ_j in (29) gives us a way of ranking the rules extracted from the knowledge base according to their relevance, which can be used to generate the recommendations that the knowledge-based innovation tutoring system will present to users.

5. Implementing a knowledge-based innovation tutoring system

In this section, we introduce the knowledge-based innovation tutoring system, describe the systemic criteria for its architecture, present the characteristics of the method for rule extraction proposed, and explain how this new method contributes to achieving some of the key systemic criteria for the architecture of the knowledge-based innovation tutoring system.

5.1. Motivation

Different innovation management tools have been proposed to deal with the processes of managing innovation and they have been implemented using conventional workflow management tools [34]. The key differentiator of the approach advocated in this article is that ours is a knowledge-based approach that guides inexperienced users through the complex process of selecting what rules to use in order for them to manage their innovation deals. While conventional approaches to innovation management let users instantiate their rules in the innovation management system as they see fit, the knowledge-based innovation tutoring system we are advocating will issue recommendations to the user as to what rules to use regardless of whether they use a conventional innovation management system to manage their innovation deals or whether they manage their deals manually or with the use of *ad hoc* management tools.

The distinction we make between conventional innovation management tools and those advocated in this article is that the former do not pursue a knowledge-based approach, that is, they all presuppose that their users are knowledgeable and experienced innovation managers who do not require expert knowledge, whether provided by human experts or by an expert system. Unlike these conventional software management tools for innovation portfolio management, product marketing, and product development management, our main focus is not on the workflow management functionality *per se*, as this area is already well-covered by several software management tools in the market [34]. All these tools assume that the user knows the knowledge rules to conduct innovation processes and therefore provide client applications allowing users to instantiate “their rules” in the system. Once instantiated the workflow management tool applies these rules in order to guide the innovation process through the innovation life cycle. Unlike these approaches that assume that users, and the organizations to which they belong, have already the knowledge required to instantiate their rules, we are addressing a segment of entrepreneurs and managers of start-ups and small and medium-sized firms arising out of the innovation networks of emerging regions of technology innovation and entrepreneurship where this knowledge is not readily available and the need for such a knowledge-based approach toward evidence-based innovation management arises. Such is the case of innovation managers in knowledge-intensive industries embedded in emerging regions of innovation and entrepreneurship as opposed to those embedded in complex innovation networks [31,71].

5.2. Systemic criteria

In this subsection, we introduce the systemic criteria for the architecture of the knowledge-based innovation tutoring system.

5.2.1. Usability

A first systemic criterion is usability. Usability has been addressed in this architecture by proposing a completely outsourced delivery model whereby users can access the functionality of the knowledge-based tutoring system via a Web browser. This allows for ease of use and spares users the complexities of installation. It also allows for the tool to be deployed using a software-as-a-service delivery model. The usability requirement also requires an easy-to-follow natural language interface to input user queries at the presentation tier.

5.2.2. Portability

Portability is addressed in this architecture by using a software-as-a-service delivery model. The presentation tier consists of a zero-client application that shall support all existing browsers and allow users to access all business process functionality at the second tier over the Web.

5.2.3. Scalability

Scalability is also an important systemic criterion for the architecture. Initially, we do not expect the system to be deployed for a large number of concurrent users. For the initial prototype scalability will not pose a concern. Once deployed as a commercial-grade application, the tool should scale to hundreds of concurrent users per second. As far as scalability is concerned for the architecture proposed, the only potential bottleneck could become the knowledge base.

5.2.4. High availability

As in the case of the scalability criterion, the only potential bottleneck as far as high availability is concerned is again the knowledge base. High-availability issues may arise whenever inferential processes are triggered in the knowledge base for the system to produce a recommendation. Our initial approach to circumventing this problem is to rely as much as possible on already-stored rules in the knowledge base and to perform updates to the rules off-line on a semi-automated basis. This process shall be conducted with the participation of human experts. Given that this application does not rely on “deep” understanding on the fly, the need for deep inferential processes to be triggered on the fly in the knowledge base shall not arise so that a lightweight knowledge rule extraction approach becomes feasible for the current implementation. Indeed, the multicriteria decision analysis approach to rule extraction described in Section 4 allows us to implement an efficient, lightweight algorithm that is efficient enough to comply with this systemic criterion.

5.2.5. Performance

The knowledge-based innovation tutoring system is not a mission-critical application posing highly stringent requirements in terms of response times and volume of transactions per second. Insofar as stored knowledge is accessed in the knowledge base, there should be no performance issues. In general, online inferential processes could potentially pose such issues, though. As our application domain does not call for heavy-duty, deep inferential processes to be triggered on the fly in the knowledge base, we do not expect the knowledge base to become a bottleneck in the architecture proposed that could compromise the performance of the knowledge-based innovation tutoring system proposed.

5.2.6. Reliability

The reliability criterion can be understood as how reliable the recommendations are, that is, whether the knowledge-based tutoring system is capable of producing a recommendation given a query posed by the user and whether the recommendation delivered by the system is relevant or not to the user (and to a human expert). Reliability will largely depend on the amount and quality of the knowledge stored in the knowledge base, on the one hand, and the ability of the system to locate and retrieve relevant rules efficiently. Therefore, satisfying this systemic criterion will very much depend on our ability to build and maintain knowledge bases that contain sufficiently large sets of validated world and application domain knowledge. Complying with this criterion will also require the implementation of efficient methods for the automated extraction of rules such as the one presented in Section 4 based on relevance metrics such as those postulated in Section 3.

In what follows, we present the architecture of the knowledge-based innovation tutoring system. As we shall see later on in this section, our choice of a multicriteria decision analysis method has been motivated by the need to comply with the systemic criteria described above, in general, and with the reliability criterion, in particular.

5.3. The architecture

Based on the systemic criteria identified for the knowledge-based innovation tutoring system, a three-tier architecture has been proposed for the first prototype of the system. The first tier is the data tier consisting of the data containing account information of users and the knowledge base containing the world and application domain knowledge rules. The second tier is the business processes tier consisting of the business rules. These business rules are classified into two main processes that we call the first-stage and the second-stage rule extraction processes, referred to as the first-stage RE process and the second-stage RE process, respectively. The first-stage RE process is comprised of business rules that rely on conventional logical approaches to retrieving business rules from a knowledge base given a user query. The second-stage RE process implements a procedure for recommending the most relevant rules out of the set of rules that are generated as a result from the first-stage RE process. The second-stage RE process is implemented using the multicriteria decision analysis model shown in Section 3 in conjunction with the multicriteria decision analysis method shown in Section 4. The third tier is the presentation tier consisting of a zero-client (a browser) allowing users to input queries into the system and access the recommendations produced by the system directly over the Web.

5.3.1. The data tier

The data tier is comprised of the database and the knowledge base. While the database shall store all information about user accounts, the knowledge base shall store all world knowledge rules as well as rules pertaining to the generic application domain of innovation management and rules about the specific application domain, which in our case corresponds to innovation and entrepreneurship in the ICT industry sector. We presuppose that the knowledge base comes equipped with a knowledge representation system capable of retrieving, generating, and updating the knowledge rules in the knowledge base. The choice of a suitable knowledge-based system and its underlying knowledge representation language poses some challenges for the present architecture of our prototype. In order to cope with the knowledge representation requirements of our application domain, we propose the use of a knowledge representation system that exploits the representational power of first-order predicate logic [8] and allows for the representation of both generic knowledge as probabilistic conditionals [5] and contextual information construed as situations, as proposed and implemented by situation-oriented semantic and knowledge representation languages based on Situation Semantics [9,30,46,70,79–81].

5.3.2. The business process tier

The second tier of this architecture consists of the business process tier and is comprised of four main modules. A first module corresponds to a natural language interface that translates the incoming query of the user expressed in natural language into a logical form expressed in first-order predicate logic. Such logical form will correspond to either a simple rule, a conjunctive rule, a disjunctive rule, or a complex rule, as defined in Section 3. This logical form corresponds to a semantic *and* knowledge representation language suitable for querying the knowledge base. The second module corresponds to what we have termed the first-stage RE process. This process yields an initial set of knowledge rules. As per Definition 5 of the model put forth in Section 3, the outcome of the first-stage RE process corresponds to an initial set of relevant knowledge rules extracted from the knowledge base according the main relevance criterion. According to this criterion, only those rules whose antecedent is matched by the logical form associated with the user query are eligible as rules of this initial set. In general, only those rules whose antecedents trigger inferences over a given threshold of interestingness will be eligible. This interestingness threshold may be defined either by the user or by the system automatically depending on the context in which the query is posed. The third module corresponds to what we have termed the second-stage RE process. The purpose of this second-stage RE process is to select, from the initial set of knowledge rules, the rules that shall be presented to the user as recommendations. This process is implemented using the criterion of *variety*, the criterion of *informativeness*, and the criterion of *specificity* introduced by Definitions 6, 8, and 10 in Section 3. Our choice of relevance criteria has been motivated by current research in the fields of computational linguistics, natural language processing, and association rule mining [10,16,95–97,100,101]. Based on these relevance criteria, the method introduced in Section 4 builds a ranking with the most relevant rules and presents them as a recommendation to the user. The fourth module corresponds to the natural language generator. This module takes as input the most relevant rules as a semantic and knowledge representation and generates a natural language expression that corresponds to the most relevant rules in the ranking.

Besides these modules, the architecture also includes business functionality pertaining to the processes that are required to manage innovation projects. These innovation management processes are conducted by the users themselves based on the recommendations that are generated by the knowledge-based innovation tutoring system given a user query. As mentioned, these users are innovation managers with the mission to drive innovation deals through the innovation life cycle. Three concepts are central to managing these innovation processes: (i) the innovation portfolio, (ii) the innovation pipeline, and (iii) the innovation deal. While the innovation portfolio gives us an overview of all innovation deals, an innovation pipeline gives us the list of all innovation deals with a focus on the phases of the innovation life cycle and the stages of the financial life cycle through which the innovation deals in the pipeline are transitioning. Therefore, the process of managing an innovation deal can be construed as a decision-making process performed by the user that will result in moving an innovation deal backward or forward through the innovation pipeline. In so doing, users will make strategic decisions that will take their innovation deals through the different phases of the innovation life cycle, that is, from strategic alignment, incubation, and development up to and through exploitation. The system benefits from the “human reasoning” of actual innovation managers as opposed to the reasoning of a knowledge-based system in the traditional sense. Users who learn new rules as they input their queries into the system will be able to judge the recommendations of the knowledge-based innovation tutoring system and learn about how to manage innovation in the process. A recommendation in this context is a series of rules the user is advised to consider as business rules regarding an innovation deal in an innovation pipeline. For each

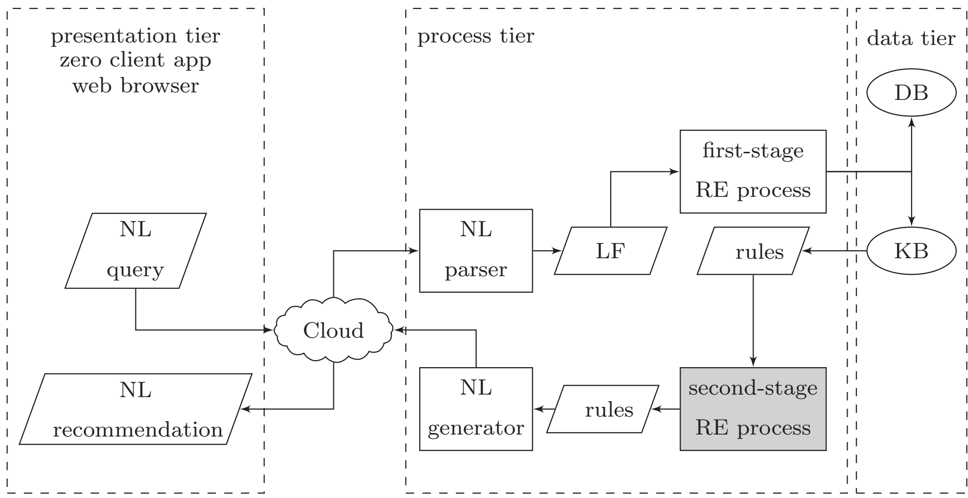

Fig. 5. The 3-tier architecture of the knowledge-based innovation tutoring system.

innovation deal in the innovation pipeline “situated” in a given emerging region of innovation and entrepreneurship and spatiotemporally located in a given phase of the innovation life cycle and at a given stage of the financial life cycle, these actions might advise the user to continue to perform certain actions in the present phase, or to go back to previous activities in the present phase or even in a previous phase of the innovation life cycle in order to revisit already accomplished actions, or to move forward the innovation deal through the pipeline to the next activity in the current phase or to the next phase of the innovation life cycle. Depending upon the recommendations of the system and the decisions made by the user, an innovation deal might move forward or backward through the innovation pipeline, or it might stay in the current phase still performing some actions, or it might be put temporarily on hold, or it might get aborted.

5.3.3. The presentation tier

The presentation tier is comprised of a zero-client allowing the user to access all recommendation functionality of the system directly over the Web. As a result, the application shall be deployed from any desktop computer, thus requiring only a Web browser for the user to interact with the system. The user will be able to input his or her queries in natural language and the recommendations will be presented to him or her also in natural language.

5.3.4. The architecture of the knowledge-based innovation tutoring system

The three-tier architecture of the knowledge-based innovation tutoring system is shown in Fig. 5. In Fig. 5, the trapezoids represent inputs or outputs, the rectangles represent processes, and the ellipses represent data. In what follows, we will describe in more detail the second-stage RE process, which is shown as a grey rectangle in Fig. 5.

5.4. The second-stage RE process

We now analyze the method for rule extraction described in Sections 3 and 4. We will address four main aspects of the method, namely, how the three criteria proposed in Section 3 account for measuring relevance, how to calculate the weights associated with these criteria, how the factor of attenuation of losses ϑ is handled, and how the method described in Section 4 provides us with a much more efficient algorithm for rule extraction than all conventional methods based on logical and nonlogical approaches.

5.4.1. The relevance criteria

The choice of relevance criteria presented in Section 3 has been motivated by research in the fields of computational linguistics and natural processing. Further motivation has also been provided by recent findings in the area of association rule mining [1,43]. These findings are relevant for our present purposes because rule association mining intends to solve exactly the opposite problem we are addressing, namely, how a set of generic rules can be extracted from a database of transactions consisting of uniquely identified subsets of items taken from a predefined set of items. In our case, these rules have been extracted already by panels of experts and they have been entered in a knowledge base complying with the principles of *nonredundancy, productivity, derivability*, and *closedness* that are postulated in connection with rule associate mining in order to augment the interestingness of the rules extracted [10,16,95–97,100,101]. Therefore, our rule extraction problem can be construed as a process of generating, from an initial set of more generic and nonredundant rules, those rules that are more interesting to a user.

The issue of interestingness in rule association mining is concerned with distilling a set of generic rules from which all related association rules can be generated. Therefore, a rule is more interesting to the extent that it is more generic and contributes to generating other observed association rules that may otherwise not be possible to generate. Interestingness in our application domain is not concerned with the process of distilling generic rules but rather with the process of discharging or generating from an initial set of generic rules those rules that contribute to answering the user query. The rules to be generated can be classified in three main categories. A first category is concerned with presenting to the user those rules that are more “generative.” This initial criterion relates to the derivability criterion of rule association mining [16] and means that a rule is more interesting to the user to the extent that we can derive more rules in the knowledge base from it. We capture this criterion through the variety function in our model of Section 3. A second category corresponds to the amount of information conveyed by the rules extracted. This second criterion relates to the productivity criterion of rule association mining in that more informative rules will be those with more correlated items in the antecedent [95]. We capture this criterion through the informativeness function in our model of Section 3. Finally, the specificity criterion provides us with a criterion that intends to deliver more specific information along competing inference chains to answer a user query. According to the specificity criterion, the longer the inference chain over a given threshold α, the more specific and hence the more relevant a rule will be.

5.4.2. Calculating the weights

The weights associated with the criteria depend on the evaluator(s). A usual approach to calculating these weights has consisted in applying a pairwise comparison process along the entire space of criteria involving the groups of evaluators that will have to deal with the concrete decision-making problem within an organization using scales such as those proposed by Saaty [76,77]. These approaches would ask evaluators to provide judgments or priorities for pairs of criteria in order to elicit the actual weights to be attached to each criterion. We do not describe this method here but refer the reader to [76,77] for a detailed description of how these weights can be elicited by asking the evaluators involved to provide their preferences or priorities regarding criteria using a pairwise comparison process and a predefined Saaty scale.

The method introduced in Section 4 proposes a more straightforward approach to calculating the weights of these criteria. This method does not deploy Saaty scales but rather asks the evaluator to provide an “importance” for each criterion in a scale from, say, 1 to 10. The method then defines the reference criterion as the one with the highest importance. Once this criterion has been fixed, the weights can be elicited using Eq. (26) in Section 4. In our knowledge-based innovation tutoring system, we let users enter the importance of each criterion along with their query. In order for them to do so, users will be advised: (i) that the variety criterion should have a higher importance whenever they are in exploratory mode, that is, when they are seeking to explore the different possible courses of action that can be followed with an innovation deal in a given phase of the innovation life cycle and at a given stage of the financial life cycle, (ii) that the informativeness criterion should have a higher importance whenever they are seeking more informative rules, that is, when they are seeking those rules that will convey more information, and (iii) that the specificity criterion should have a higher importance whenever they are seeking a concrete answer to a very specific question. With this information in place, the system can assign the weight of each criterion using Eq. (26).

5.4.3. Setting up the factor of attenuation of losses

As mentioned in Section 4, the use of TODIM as a multicriteria decision analysis method has been motivated by the need to provide a psychological account of decision-making under risk, as provided by Prospect Theory and Cumulative Prospect Theory [48,91]. Although some authors might argue that there is no reason why a decision-aiding system should deploy psychologically or cognitively adequate algorithms that mirror the way humans make decisions under risk, we argue that for many application domains this should indeed be a requirement. In the case of innovation, this is particularly important due to the inherent uncertainty associated with processes of innovation and entrepreneurship. In fact, there is enough empirical evidence demonstrating that companies make strategic investment decisions regarding innovation, especially radical and substantial innovation, not based on the future asset position of the company but on their perception of gains or losses based on the current status quo. The most common expression of this phenomenon is the well-understood dilemma of creative destruction, also referred to as the innovator’s dilemma [22].

As mentioned in Section 4, the value function of TODIM accounts for three main psychological processes of human decision makers, namely, the fact that the value function in human decision-making under risk adopts: (i) a concave form showing the aversion to risk of decision makers when they experience gains, (ii) a convex form showing the aversion to losses in its convex (negative) part when losses arise, and (iii) a steeper form in its convex (negative) part showing the risk-seeking behavior of decision makers when losses arise. This third component of the value function of TODIM is accounted for by way of introducing the factor of attenuation of losses ϑ. When $\vartheta = 1$, the value function of TODIM in its convex (negative) part does not entail any propensity to risk at all. For values of $\vartheta > 1$, this factor of attenuation of losses leads to a factor $\lambda = 1/\vartheta$ of loss aversion < 1, as predicted by Prospect Theory and Cumulative Prospect Theory.

In the multicriteria decision analysis literature, the factor of attenuation of losses is not calculated but rather set up at a certain value, or range of values, for the decision makers involved, who in our particular case correspond to innovation managers. Therefore, ϑ is construed as a parameter in TODIM that is to be set up by assuming a certain degree of propensity to risk on the part of decision makers when losses arise. It is also common practice in the multicriteria decision analysis literature to conduct sensitivity analyses using this parameter in order to ascertain what range of values of ϑ would turn out to be order-preserving. The treatment of ϑ in the multicriteria decision analysis literature as a parameter that needs to be set up and not calculated is based on the less practical use of calculating its value from a decision theory standpoint. Indeed, most of the studies that have attempted to elicit this and other factors affecting the shape of the so-called probability weighting function, a function that shapes the value function of decision-making under risk in humans, have come from the cognitive science research community. These studies have all dealt with basic research aimed at understanding the psychophysics of decision-making under risk in humans. The values of these parameters are a function of four main variables and they are elicited by designing experiments and analyzing the resulting data using nonlinear regression. The variables involved in eliciting the values of these parameters correspond to: (i) the given decision-making task; (ii) the given presentation mode; (iii) the decision maker; and (iv) the given application domain [17,41].

In most applications, it would be rather impractical to calculate the values of these parameters, including the value of ϑ, using the values of the four variables mentioned above. Such an undertaking would make little practical sense for multicriteria decision analysis applications because in any decision-making situation, there would be too many potential different values for these parameters depending on the different combinations of values for the variables mentioned above. This has led to a more pragmatic approach to incorporating the results of Prospect Theory and Cumulative Prospect Theory [48,91] into multicriteria decision analysis by construing these values as parameters that need to be set up *a priori* and by conducting sensitivity analyses aimed at ascertaining what ranges of values would preserve the rankings or orderings obtained. Following such an approach, the value of ϑ is set up at a predefined value, or range of values, assuming different degrees of propensity to risk on the part of evaluators.

As a quintessential case of applying an experimental research methodology, the psychophysical studies that led to the formulation and validation of Prospect Theory and Cumulative Prospect Theory have been highly relevant in that they have contributed to our understanding of how human decision-making under risk deviates considerably from the original precepts of utility theory [69]. As such, the aim of these empirical studies has not been to put forward a general methodology for eliciting the “actual value of these parameters” in a given application domain in order to apply such values in an actual decision-aiding task. These empirical studies have not aimed to elicit “values” for these parameters that can be used as constants across application domains either, as such values will vary in every decision-making situation under risk depending on the values of the four variables mentioned above. The main goal of these studies was to understand the psychophysics of human decision-making under risk. These studies have called into question the adequacy of utility theory as a framework for decision making under risk and have demonstrated:

(i) that humans tend to make decisions under risk not based on final asset positions but rather on gains and losses that are measured using current reference points, (ii) that humans show a general aversion to risk when gains arise, and (iii) that they show a general propensity to risk – and hence to loss aversion – when losses arise.

5.4.4. The complexity of the method

As mentioned in Section 1, the method for rule extraction presented in this article has been designed to provide an efficient way to extract relevant rules from a knowledge base to be used in concert with a knowledge-based innovation tutoring system. The method builds upon the algorithm described in Section 4 for building a ranking of relevance. As this algorithm is based on the pairwise comparison of alternatives built by the function ϕ, it follows that its complexity corresponds to $\mathcal{O}(n^2)$, where n is the number of alternatives extracted from the first-stage RE process shown in Fig. 5. Thus, the method not only outperforms the most efficient rule extraction algorithms proposed by symbolic, subsymbolic, and hybrid rule extraction approaches but also the frequent association rule mining approaches described in Section 2.

The reason for this gain in efficiency is based on the complexity reduction obtained by the deployment of the "intelligence" of users in the process of actually building final rankings of rules and learning what rules to apply and when to apply them. This complexity reduction is granted in knowledge-based tutoring systems such as the one proposed here to solve the knowledge bottleneck problem of innovation management for reasons that we discuss in Section 6. The complexity of the functions *variety* and *informativeness* is $\theta(n)$, which makes the computation of these two criteria computationally inexpensive. We can even tag these rules with their corresponding variety and informativeness values, thus making the retrieval of these values from the knowledge base a rather straightforward process.

The calculation of the specificity criterion is more computationally challenging as its value needs to be computed in real time and can involve the exploration of parallel inference chains, thus making the computation of the function *specificity* potentially intractable. There are several ways to deal with this problem. The most obvious one is to prune the search space of more specific rules by choosing a suitable value for the "interestingness" threshold. But even if we propose a low threshold such as 0.25, the search space could still pose some challenges in many cases, as shown by Example 9 in Section 3.

Our approach to reducing the complexity of this function will depart from considering an interestingness threshold as the only parameter, though. We will also assign a maximum value of specificity that can be set up by the user and will typically not exceed the value 3. Thus, following Example 9, it would suffice to explore the inference chain $r_1 : a_1 \longrightarrow^{p_1=0.384} (c_{1.1} \wedge c_{1.1.1} \wedge c_{1.1.1.2})$ in order to ascertain that the specificity of r_1 has at least the value 3. This requires a slightly different definition of the specificity function. In this particular case, this would mean that the specificity of a rule can range from 0 to 3. In other words, rules that carry a specificity value of at least 3 would be considered specific enough to get the maximum value under the specificity criterion. Furthermore, the weight of the specificity criterion can be set to zero in cases where the user is not in "exploitation" mode, that is, in cases in which the user has posed a query that does not require the system to find an answer along a specific inference chain. In such cases, the user would be in an "exploratory" mode and the method proposed to rank the alternatives would turn out to be extremely efficient, as mentioned above. But even when the user is in an exploitation mode, our task does not consist in searching the entire space of inference chains in search of an actual answer to the user query along competing inference chains but rather in building a ranking of rules that gives a higher weight to the specificity criterion. Our method would position more specific rules higher in the ranking and it would be the task of the user to choose and apply one of the recommended rules in the knowledge-based innovation tutoring system in order to solve the task at hand and learn from this process.

We therefore conclude that the complexity of the method proposed in the cases in which the specificity criterion is of the essence will be dramatically reduced by our implementation because an exhaustive search over the entire inference space is not required. Assuming that we implement the function *specificity* using a depth-bounded, depth-first search, at a depth (specificity) k and assuming a maximum branching factor b for the complex rules involved, the complexity of the *specificity* function is $\mathcal{O}(b^k)$. As mentioned, if we limit the upper bound of the specificity criterion to $k = 3$, the complexity of the *specificity* function is $\mathcal{O}(b^3)$, thus providing a highly efficient upper bound for the time complexity of the *specificity* function. For suitable values of b and n such that $b \ll n$ in our complex rules, the total upper-bound time complexity of our second-stage RE process will still be n^2 even in cases where the user is in exploitation mode and the specificity function needs to be invoked.

In a recent contribution by Webb and Vreeken [97], a highly efficient algorithm for extracting key associations from self-sufficient itemsets was presented. The OPUS Miner algorithm proposed solves a problem similar to the one described in this article by applying a depth-first search of self-sufficient itemsets using efficient pruning mechanisms. With a time complexity of $\mathcal{O}(2^n)$, where n is the number of items in the database, this algorithm is much more efficient than other efficient algorithms for frequent association rule mining [11,93,94,98]. Its time complexity shows that the worst-case scenario for rule extraction using this algorithm is clearly intractable. This is to be compared with our algorithm, whose time complexity of $\mathcal{O}(n^2)$, where n is the number of initial rules extracted from the knowledge base, shows that n^2 provides an upper bound of time complexity. Thus, the algorithm we have proposed to implement our second-stage RE process is much more efficient when compared with the most efficient algorithms for rule extraction proposed in the knowledge discovery and data mining research fields, all of which are not based on multicriteria decision analysis methods but on conventional logical or nonlogical approaches.

6. Discussion

Differently from other software tools for innovation portfolio management, product marketing management, and product development management, our main focus in building a knowledge-based innovation tutoring system is not on the workflow management functionality *per se*, as this area is already well-served by enterprise software applications whose functionality is based on conventional workflow management systems [34]. Unfortunately, all these management systems assume that the user is an experienced innovation manager who already "knows" what rules to apply in order to manage complex innovation processes. These systems assume that users are able to instantiate the proper rules in the system as the development of products or services progresses along the entire innovation life cycle. Therefore, these tools limit themselves to providing client applications allowing users to instantiate their rules in the system in a way that is comprehensible and easy to use. Once instantiated, the workflow management tool applies these rules in order to guide the innovation process through the entire innovation life cycle.

Unlike conventional approaches not based on knowledge-based management systems, we are addressing a segment of entrepreneurs and managers of start-ups and small and medium-sized firms in emerging regions of technology innovation and entrepreneurship where this knowledge is not readily available and the need for such knowledge-based approaches toward innovation management arises. The knowledge-based innovation management system advocated in this article is intended to assist a user in making their own decisions about what rules to apply based on the recommendations presented by the system. Following such recommendations, the user can apply the rules in the knowledge-based system as they see fit. The method for automatic rule extraction presented in this article allows us to implement a solution to the knowledge extraction problem presented in Section 3. The solution proposed is based on mapping the problem of rule extraction from large knowledge bases to a multicriteria decision analysis problem.

The multicriteria decision analysis method presented in Section 4 has been chosen with a view to implementing an efficient algorithm for rule extraction in the system. The two-step method for rule extraction introduced constitutes the cornerstone of the approach presented in this article. This method uses an initial step based on conventional logical approaches to retrieve an initial set of rules. Here, we rely on the standard functionality of conventional knowledge representation systems such as those surveyed in Section 2. The second step, though, is quite a departure from the standard symbolic, subsymbolic, and hybrid approaches in that we introduce a set of relevance criteria to rank the initial set of rules using multicriteria decision analysis. We are able to do so because the main goal of the knowledge-based system proposed is not to build a system for automatic deep reasoning, as is usually the case in knowledge-based applications [80,88], but rather to build an automatic recommendation system that assists users in choosing and learning the proper innovation management rules and letting them use these rules to manage their innovation management processes through the innovation life cycle. Thus, our approach relies not only on artificial intelligence provided by the system but also on the intelligence and reasoning of users. As a by-product, users learn what rules to deploy based on the context of the queries they pose and the rules recommended by the system.

The criteria chosen to implement the ranking algorithm has been based on the concept of relevance. For the selection of the relevance criteria presented in Section 3, we have relied on previous work in the area of knowledge representation and knowledge-based representation systems, on the one hand, and frequent association rule mining, on the other, which we surveyed in Section 2 and then elaborated upon in Section 5. As far as the implementation of the system is concerned, the variety and informativeness criteria shall not pose a threat to scalability and performance of the system since the calculation of the value for these criteria is rather straightforward. Furthermore, the calculation may be facilitated if the rules are annotated with metadata containing their values of variety and informativeness. The careful knowledge engineer should write the rules making sure that this information is hardcoded as metadata while entering the rules in the knowledge base. The specificity criterion may pose some challenges, though, as the calculation of its value will depend on the number of inferences triggered once a query q is asserted in the knowledge base and on the interestingness threshold α.

As mentioned, the computational effort required for this calculation can grow dramatically as the number of inferences triggered in the knowledge base grows, even for relatively high values of the interestingness threshold α. We have chosen a pruning method to circumvent this problem based on a depth-bounded, depth-first search method where both the depth and the interestingness threshold can be set up either by the system or the user. This would result in avoiding the calculation of path lengths for paths that are either below the interestingness threshold or beyond the specified depth. As mentioned, the function *specificity* would only be invoked in cases where this criterion matters. For instance, in case the query q posed by the user were generic or underspecified, it would be reasonable to assume that the user is in “exploratory mode.” In such cases, the variety criterion should take precedence over the informativeness and the specificity criteria. In fact, the specificity criterion may be dropped altogether in some of these cases. In cases where the query q were more specific, as in the case of a more complex query matching an antecedent involving more attributes with their respective values, it would be reasonable to assume that the user is in “exploitation mode.” In such cases, the search for more specific information along an inferential path would yield more specific information and the specificity criterion should take precedence over the variety criterion. In the multicriteria decision analysis method introduced in Section 4, we can accomplish this by choosing the weights of all criteria accordingly.

From a multicriteria decision analysis standpoint, there is a very interesting issue that arises in our application domain. Rule extraction from large knowledge bases is more singular than other application domains in that the relative weights associated with the set of criteria C can change dynamically depending on the query q entered by the user. If the query is rather vague and underspecified, then the variety criterion should carry a higher weight because in this case a broad exploration of rules in the knowledge base may lead to higher value to the user. On the other hand, if the query q is very well specified, then the specificity criterion should carry a higher weight than the other two criteria. Therefore, weights can be automatically modified in the system depending on the query q entered by the user. A user familiar with the system and the way the rules are coded might also provide this information as part of the query.

Finally, our application domain also provides a good example of applications in which the “intelligence” of the user can be deployed in the system and can contribute to reducing the complexity of the algorithms involved. This is particularly the case in knowledge-based tutoring systems such as the one presented in this article, where the user is expected: (i) to choose the most relevant rules from a final recommendation of the system presented in the form of a ranking of rules based on relevance, (ii) to apply them for innovation management purposes, and (iii) to learn from this process.

7. Conclusions

The work reported in this article covers uncharted territory at the intersection of knowledge-based systems in the artificial intelligence tradition, innovation management, and multicriteria decision analysis. From both a theoretical and practical standpoint, the use of multicriteria decision analysis methods is a novelty in the knowledge-based systems community, a community whose main focus has traditionally been on proposing and implementing knowledge representation languages and systems with a view to endowing them with automated inferential capabilities for either shallow or deep reasoning. In this article, we have demonstrated how the problem of rule extraction from large knowledge bases can be defined as a multicriteria decision analysis problem. We have put forth a multicriteria decision analysis method for implementing the ranking algorithm required to order the rules extracted from the knowledge base according to their relevance. We have also provided a fundamental proof that shows that our method outperforms the most efficient algorithms for rule extraction proposed in the association rule mining and knowledge discovery research communities [1,50,61,97]. Our proof is based on the theoretical complexity of the algorithm proposed, which we have also compared with the theoretical complexity of what can be regarded as one of the most efficient algorithms for rule extraction proposed to date [97].

The method for rule extraction presented in Section 4 allows us to implement a ranking algorithm that can be used for automatic rule extraction in knowledge-based systems for other application domains as well. Differently from other domains that may use state-of-the-art automated knowledge acquisition tools and methods for acquiring knowledge rules and factoids from structured and unstructured data sources such as the Internet [24–26,42,79,80,82], our application domain relies more on experts and practitioners who play an active role in diffusing knowledge in emerging and complex innovation networks in knowledge-intensive industries. In this connection, our endeavor of building knowledge-based innovation management systems will require the concerted effort of these actors in order to build knowledge bases large enough for commercial-grade applications.

Due to the novelty of both the rule extraction method and the application presented, we cannot yet provide a practical comparison or benchmark of the method proposed against other cases. This shall remain the subject of future work. Given the advantages in terms of the theoretical complexity of the rule extraction method proposed, though, we expect that researchers and practitioners building applications such as the one put forth in this article will commence to explore the application of multicriteria decision analysis, in general, and the method proposed here, in particular, to build relevance rankings more efficiently for the purpose of rule extraction in knowledge-based systems.

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